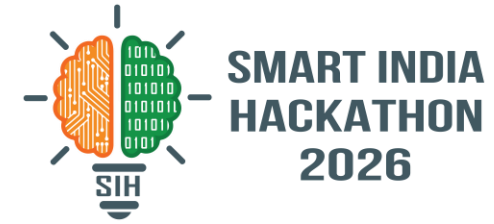
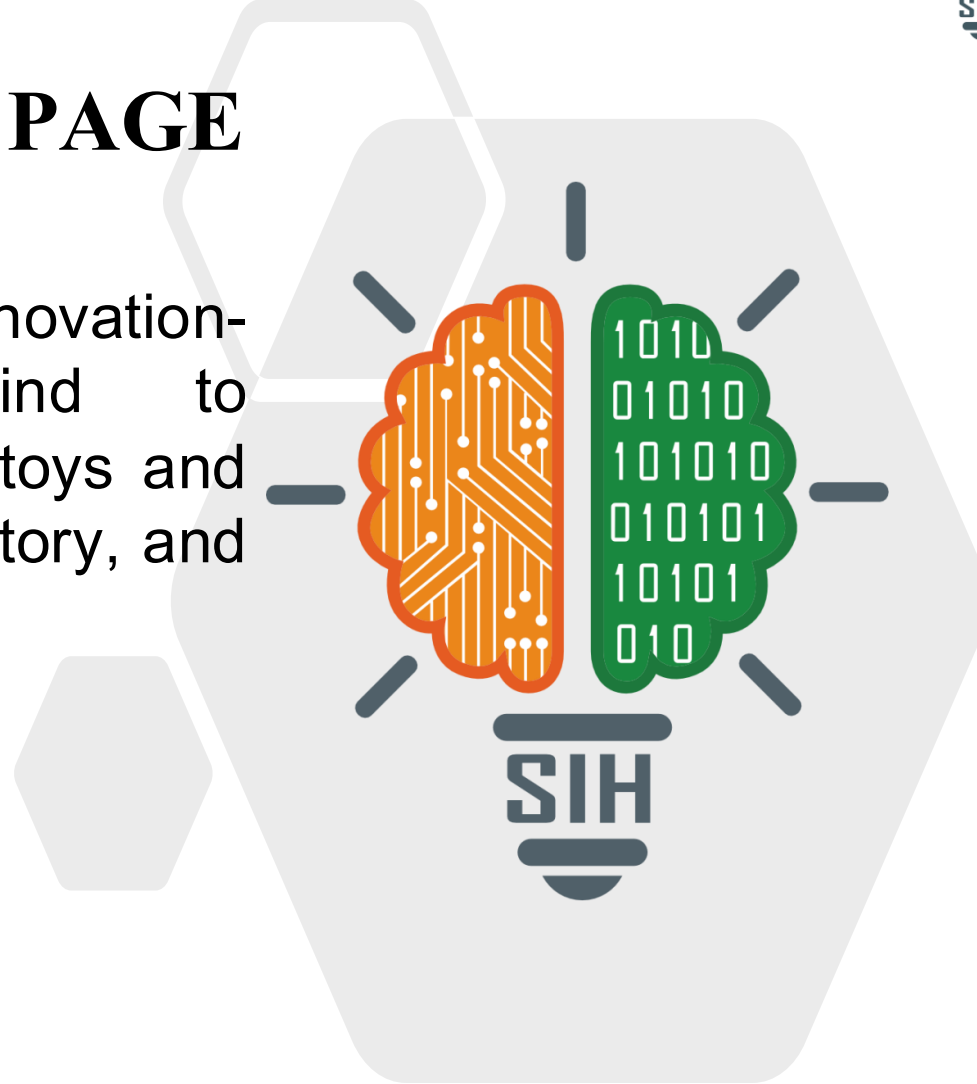


# SMART INDIA HACKATHON 2026



## TITLE PAGE

- **Problem Statement ID** – SIH26208
- **Problem Statement Title**- Student Innovation- Challenge your creative mind to conceptualize and develop unique toys and games based on our civilization, history, and culture etc.
- **Theme**- Toys & Games
- **PS Category**- Software
- **Team ID**- 126110
- **Team Name** : Solution X



## Idea / Solution

- **Build an immersive, story-driven game** that transforms selected cultural, traditional and historical narratives into interactive experiences.
- **Create four themed worlds — Cosmology, Ancient Bharat, British India and Present India** — offering a progressive journey through selected cultural and historical narratives.
- **Integrate 60–90 second cinematic storytelling with gameplay**, followed by exploration, puzzles, challenges and knowledge-based quizzes.
- Use scores, rewards and achievements to drive game progression and engagement

## Gameplay Flow

 Story →  Explore →  Interact →  Puzzle →  
 Challenge →  Reward →  Progress

## Problem Resolution

- **Transforms passive learning into interactive gameplay** through stories, exploration, puzzles, and quizzes.
- **Boosts engagement and understanding** by connecting storytelling with gameplay challenges.
- **Offers an extensible framework** for adding new regions, traditions, languages, history, and levels.

## Unique Value Propositions (UVP)

- **Narrative-to-Gameplay:** Stories directly drive puzzles, challenges, and quizzes.
- **Four-World Cultural choices :** Four themed worlds choices mythology/traditional narratives, ancient civilization, British India and contemporary India within one gaming framework.
- **Learn by Playing:** Progression, rewards, and challenges make cultural learning interactive.



## Story → Gameplay

Cultural narratives directly contextualize puzzles, quest lines, and tactical challenges.

## Four Independent Cultural Worlds

Players choose from four distinct cultural experiences — Cosmology, Ancient Bharat, British India, and Present India — within one expandable game framework.

## Learn Through Discovery

Instead of simply answering questions about heritage, players discover knowledge by exploring environments, solving challenges, and interacting with the world.

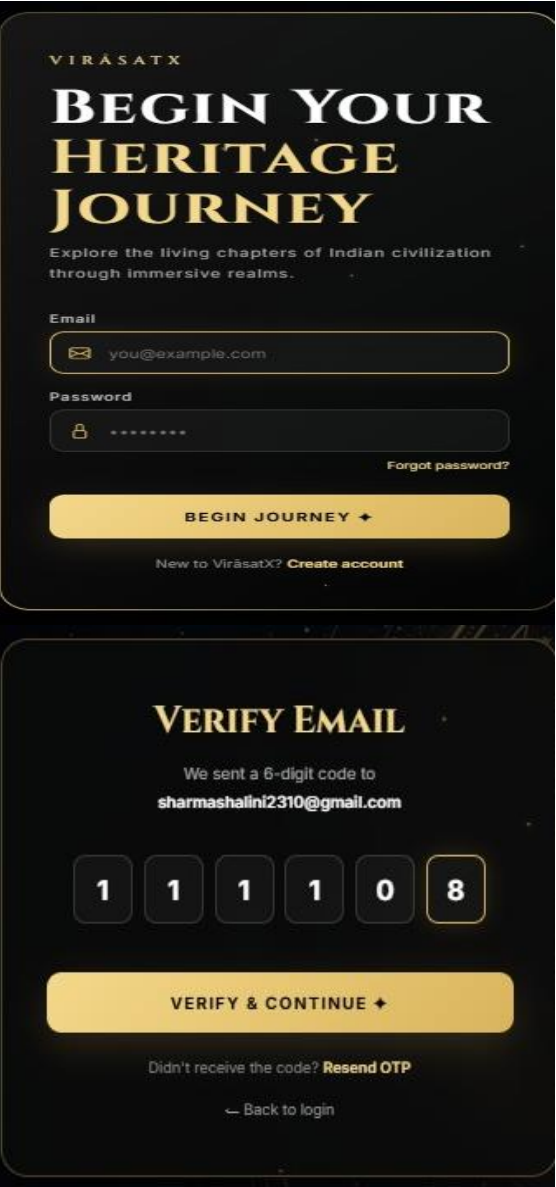
**HERITAGE → STORY  
→ GAMEPLAY**

## Cinematic Cultural Context

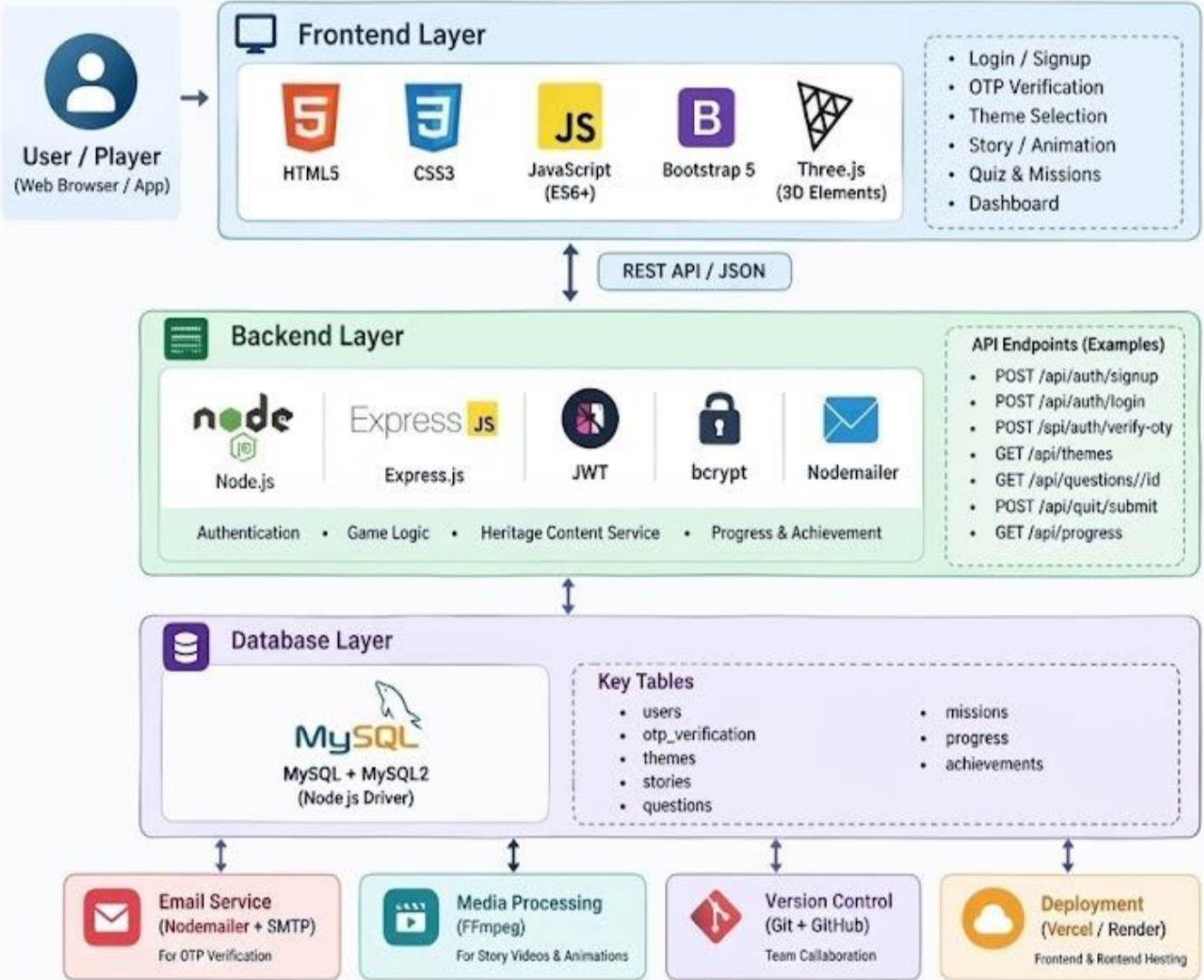
Short story-driven cinematic sequences introduce each theme and create context before gameplay begins, turning historical and cultural content into an experience.

## Expandable Heritage Framework

The same game framework can grow with new regions, traditions, languages, historical periods, stories, puzzles, and levels.



Prototype Email Verification



## Technology Used

### Frontend

- HTML5
- CSS3
- JavaScript (ES6+)
- Bootstrap 5
- Three.js (3D Elements)

### Backend

- Node.js
- Express.js
- REST API
- JSON

### Database

- MySQL
- MySQL2

### Security & Authentication

- JWT
- bcrypt
- Nodemailer
- OTP Verification

### Game & Media

- JavaScript Game Logic
- FFmpeg
- Interactive 3D Elements

### Development & Deployment

- Git
- GitHub
- VS Code
- Vercel / Render

● Implemented ● In Progress ● Planned

## Feasibility

- **Open-source & affordable:** HTML, CSS, JS, Node.js, MySQL
- **Modular architecture:** Easily add themes, stories, quizzes & missions
- **Cross-device:** Responsive and lightweight web application
- **Content validation:** Reliable sources + manual verification

## Viability

- **Educational impact:** Storytelling + quizzes → active learning
- **Target users:** Students, educators & culture enthusiasts
- **Low-cost:** Open-source stack + scalable deployment
- **Future-ready:** New themes, languages & mobile expansion



## Potential Challenges → Proposed Solutions



### Large Media Files

High-resolution videos and images can increase load time and storage usage.



### Compression & Optimization

Use efficient formats (WebP, MP4) and compress media files.



### Low Bandwidth / Internet

Slow or unstable internet can affect user experience and media loading.



### Caching & Lightweight Assets

Use caching and lightweight assets for smoother performance.



### Device Limitations

Different devices have varying screen sizes, processing power and browsers.



### Responsive Design + Fallback

Use flexible layouts and fallback content for older devices.



### Content Accuracy

Risk of incorrect or biased cultural/historical information.



### Source Verification + Review

Use reliable references and manual review before publishing content.



### Increasing Content / Scalability

More themes, stories and users can increase complexity and database load.



### Modular Architecture

Use a modular structure with separate services and scalable database design.

# IMPACT AND BENEFITS

## Interactive Learning

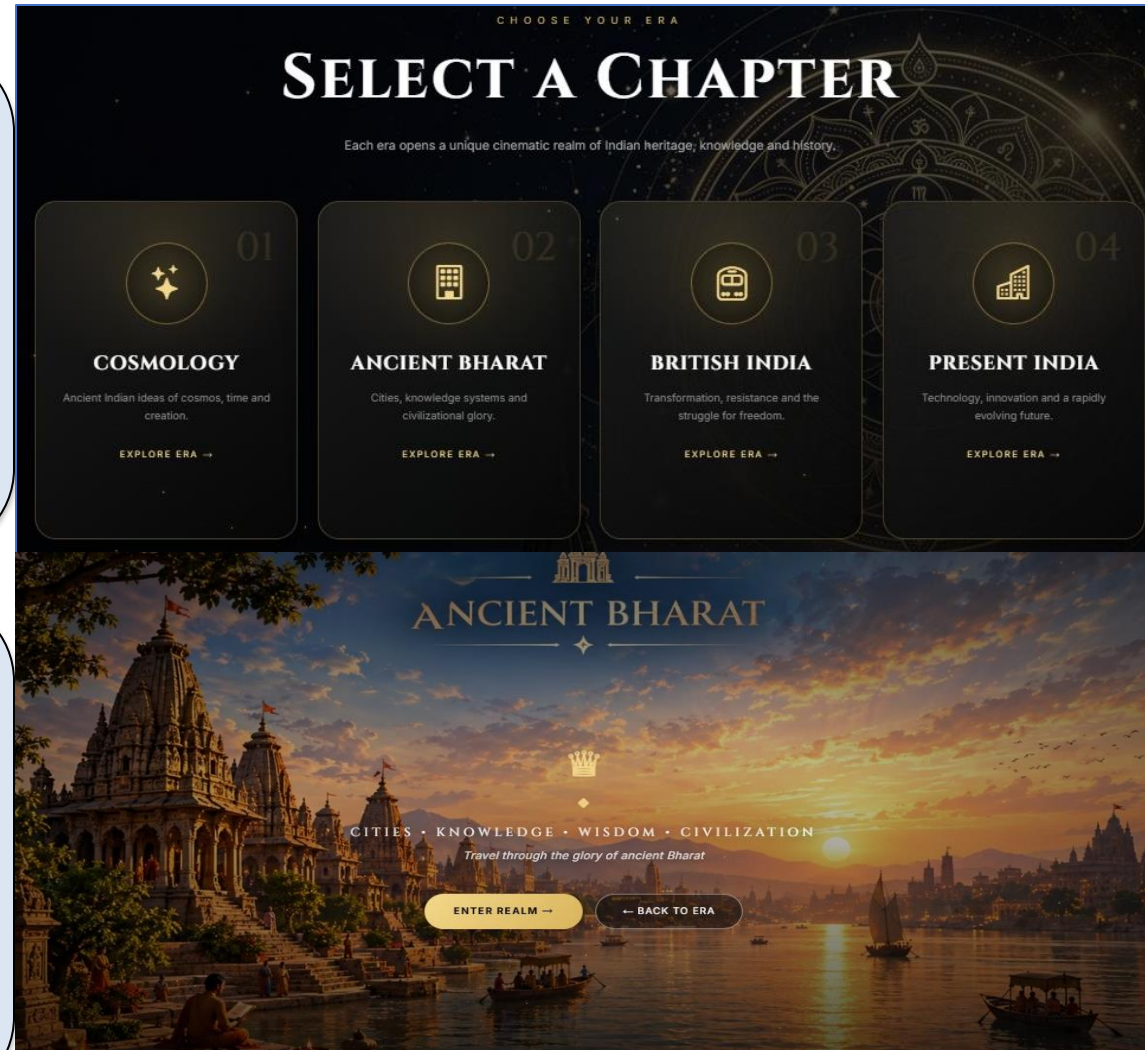


Transforms cultural education into an engaging experience through storytelling, puzzles, quizzes, and exploration.

## Scalable Opportunity



Creates a scalable platform for educational gaming across schools, tourism, museums, and digital learning.



Prototype cultural connection and Heritage diversity

## Cultural Connection



Builds cultural awareness by connecting players with India's diverse heritage, traditions, and history.

## Digital Heritage



Enables immersive exploration of cultural heritage through a digital, resource-efficient experience.

| Area                      | Reference                                                                                        | Link                                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| GitHub                    | <b>Prototype Deployed on GitHub</b> -This is the URL of our project which is Deployed on GitHub. | <a href="https://github.com/sharmashalini2310-source/-VirasatX-">https://github.com/sharmashalini2310-source/-VirasatX-</a> |
| 3D Characters & Assets    | <b>Mixamo</b> – character models, rigging & animations                                           | <a href="https://www.mixamo.com/">https://www.mixamo.com/</a>                                                               |
| Weapons & Costumes        | <b>Sketchfab</b> – 3D references/assets for weapons, clothing & environments                     | <a href="https://sketchfab.com/">https://sketchfab.com/</a>                                                                 |
| Audio / Narration         | <b>ElevenLabs</b> – AI voice/narration reference for cinematic storytelling                      | <a href="https://elevenlabs.io/">https://elevenlabs.io/</a>                                                                 |
| Cinematic Sequences       | <b>Motion Array</b> – cinematic templates, visual references & storytelling assets               | <a href="https://motionarray.com/">https://motionarray.com/</a>                                                             |
| 3D Design & Development   | <b>Blender</b> – modelling, scene design, character/environment preparation                      | <a href="https://www.blender.org/">https://www.blender.org/</a>                                                             |
| Cultural Game Inspiration | <b>Raji: An Ancient Epic</b> – Indian mythology, environments, characters & visual storytelling  | <a href="https://rajithegame.com/">https://rajithegame.com/</a>                                                             |